

Threat Intelligence Report: Kinsing Botnet Campaign Detection & Attribution

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Project:	The Littlefish Hunter – AI-Powered Honeypot & Threat Detection Node
Target Host:	rpilab (Cowrie Honeypot deployed in Docker Container)
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1. Executive Summary

Within the framework of the automated threat monitoring infrastructure, the Wazuh SIEM system detected a sustained, high-frequency SSH brute-force attack targeting the **rpilab** environment. To capture adversary tactics, techniques, and procedures (TTPs), the malicious traffic was seamlessly funneled into an interactive Cowrie honeypot. This report details the forensic collection of Indicators of Compromise (IoCs), raw log analysis from the Docker container, and deep-dive attribution using external Cyber Threat Intelligence (CTI) feeds including OSINT, VirusTotal, and GreyNoise.

2. Forensic Log Analysis & Data Extraction Workflow

To extract intelligence from the active threat vector, a systematic forensic logging sequence was executed directly on the host hypervisor. The command workflow and its technical outcomes are structured below:

Phase	Executed Command	Technical Objective & Outcome
1. Container Audit	<code>docker ps</code>	Identified the active runtime state of the honeypot container. Confirmed the container name <code>cowrie</code> and established an uninterrupted uptime baseline of 9 consecutive days.
2. Log Extraction	<code>docker cp cowrie:/home/cowrie/cowrie/log/cowrie.json ./cowrie_local.json</code>	Bypassed the lack of native string-parsing binaries (e.g., <code>grep</code>) within the hardened, minimal Docker runtime container. Safely transferred a 56.9 MB raw telemetry dataset to the local host filesystem for secure offline forensics.
3. Regex Filtering	<code>grep -o "ssh-rsa [A-Za-z0-9+/*=\\?]" cowrie_local.json sort -u > botnet_key.txt</code>	Isolated structural cryptographic signatures. Used POSIX-extended regular expressions to strip out event metadata and extract unique <code>ssh-rsa</code> public key strings. The results were aggregated and pipe-sorted to strip repetitive automated hits, resulting in 29 unique botnet public keys.
4. Payload Inspection	<code>cat botnet_key.txt</code>	Rendered the contents of the structured text document to cross-reference the active payload architecture against known threat group initial access parameters.
5. Artifact Hashing	<code>sha256sum botnet_key.txt</code>	Computed a cryptographic hash unique to this cumulative 9-day ingestion signature for subsequent threat sharing and localized threat hunting exercises.

Forensic Artifact Evidence Metadata:

Parameter	Value / Structural Attribute
File Name	<code>botnet_key.txt</code>
Unique Keys Contained	29 unique cryptographic strings
SHA-256 Hash	464bdc75e2faef78d2b7818bb20894f09d8123bf0134015694c92e7072db41f5

Direct querying of the overall file hash in VirusTotal yielded a clean result (0/70). From a forensic perspective, this is a standard operational anomaly: the specific grouping of these 29 public keys represents a unique temporal signature captured exclusively by the `rpilab` node. Consequently, the composite file hash was unknown to global antivirus indexes, necessitating atomic IoC and network infrastructure tracking.

3. Deep-Dive Case Study: Infrastructure Breakdown (IP: 89.144.212.97)

To establish a verified attack profile, a principal attacking IP address extracted from the Wazuh alerts was subjected to comprehensive multi-source OSINT analysis.

Attacking Host Technical Profile:

Indicator (IoC)	Value and Intelligence Attributes
Source IP Address	89 . 144 . 212 . 97
Autonomous System	AS 8447 (Telekom Austria)
Global Reputation	Flagged explicitly as Malicious / Malware Host by GreyNoise, Cluster25, SOCRadar, and IPsum feeds.
Traffic Classification	Internet Background Noise (Mass internet-wide opportunistic vulnerability scanning)

Relationships & Passive DNS (pDNS) Graph Dissection

Querying the atomic IP address within the relational mapping interface of VirusTotal revealed critical backend data that completely reframes the nature of the threat actor's infrastructure:

- 1. Passive DNS Replication:** Historical resolution logs linked the IP 89 . 144 . 212 . 97 directly to the following active domains:
gesinase.direct.quickconnect.to
gesi1506.direct.quickconnect.to
 - **Target Device Identification:** The domain namespace quickconnect . to is the proprietary Dynamic DNS (DDNS) and relay service operated by **Synology** for consumer and enterprise Network Attached Storage (NAS) appliances.
 - **Forensic Conclusion:** The host attacking the rpilab infrastructure is not a dedicated command-and-control (C2) server provisioned by the hacker. Instead, it is a ****compromised, legitimately owned Synology NAS device**** residing within the Telekom Austria residential/commercial ISP block. The device was previously breached by a worm-like botnet, and its local processing power has been weaponized to systematically scan the IPv4 space for further targets.

4. Adversary Attribution – The Kinsing Campaign

The primary SSH key injected at the initiation of the session exhibited the following structural signature:

```
ssh-rsa AAAAB3NzaC1yc2EAAAABIwAAAQEAwMfqsxMsZ5D01HAoB1QnZXc8g...
```

Cross-referencing this cryptographic string with known malicious signature repositories provides definitive, high-confidence attribution to the ****Kinsing (Monero Crypto-Jacking Botnet)**** threat group.

Anatomy of a Kinsing Attack Lifecycle:

- **Initial Access:** Automates scanning to locate exposed services (misconfigured SSH, unauthenticated Docker API endpoints, Redis, or Elasticsearch).
- **Evasion & Cleansing:** Executes defensive evasion by purging rival botnets (commonly executing tasks like `rm -rf .ssh` to completely wipe prior authorized configurations).
- **Persistence:** Appends the specific `ssh-rsa` public key found in our `botnet_key.txt` dump directly into the local `authorized_keys` configuration.
- **Monetization (Impact):** Deploys a main ELF bot binary alongside a heavily obfuscated instance of XMRig, a high-efficiency miner designed to stealthily mine Monero (XMR), maxing out CPU utilization and exhausting local infrastructure.

5. Conclusions & Hardening Recommendations for rpilab

The successful interception, structural extraction, and behavioral analysis of this attack validate the engineering architecture of the deployed telemetry stack (Wazuh + Cowrie + Docker). To protect production counterparts from similar exploitation, the following remediation baselines are established:

- Disable password-based SSH authentication universally across the estate, enforcing strict public-key cryptography.
- Implement automated firewall blocklists using Fail2ban or active response rules within Wazuh to drop traffic from IPs showing multi-frequency authentication failures.
- Apply tight network isolation (Docker network bridging controls) and establish real-time Prometheus/Grafana alerts for anomalous CPU spikes signifying cryptojacking payloads.

Report automatically generated by "The Littlefish Hunter" CTI correlation node inside rpilab.
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